

Requirement Specification Document

Library Management Database System

Week 1 – Requirement Analysis and Customer Database Module

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Week	Week 1

1. Introduction

1.1 Purpose

The purpose of this document is to analyze and specify the business requirements of the Library Management Database System. This system is designed to manage books, members, librarians, borrowing transactions, and return records within a library. It helps maintain accurate records of books and members while ensuring efficient library operations, reducing manual work, and improving data management.

1.2 Project Scope

The Library Management Database System enables libraries to maintain records of books, members, librarians, and borrowing transactions. The system supports member registration, book catalog management, book issue and return, due date tracking, fine management, and report generation. It will be developed using a relational database with properly defined tables and relationships to ensure data consistency and efficient management.

1.3 Intended Users

- **Members** – Search books, borrow books, return books, and view borrowing history.
- **Librarians** – Manage books, members, borrowing transactions, and fine records.
- **Library Administrator** – Manage the overall database, generate reports, and monitor library activities.

2. Business Requirements

The following are the key business needs the system must satisfy:

- Maintain accurate and up-to-date member records.
- Maintain a catalog of books with title, author, publisher, ISBN, and availability status.
- Allow members to borrow and return books.
- Track the status of each book (Available, Issued, Reserved, Lost, or Under Maintenance).
- Record borrowing, return, renewal, and fine payment details.
- Generate reports such as issued books, overdue books, available books, and active members.
- Prevent duplicate member registrations by maintaining unique membership records.
- Enable librarians to efficiently manage library resources.

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
FR-1	The system shall allow new members to register with their name, email, phone number, and address.
FR-2	The system shall allow librarians to add, update, and remove books from the library catalog.
FR-3	The system shall allow members to borrow one or more available books.
FR-4	The system shall automatically record the issue date, due date, and return date for each borrowed book.
FR-5	The system shall calculate overdue fines automatically when books are returned after the due date.
FR-6	The system shall allow librarians to update the status of books (Available, Issued, Reserved, Lost, or Under Maintenance).
FR-7	The system shall allow members to view their borrowing history and currently issued books.
FR-8	The system shall allow administrators to generate reports such as issued books, overdue books, available books, and member activity.

4. Non-Functional Requirements

- **Performance:** The system should retrieve book and member details within 2 seconds.
- **Security:** Member information and librarian credentials must be stored securely, and only authorized users should access library records.
- **Reliability:** The database must maintain data consistency by ensuring that a book cannot be issued to multiple members simultaneously.
- **Usability:** The user interface should be simple and easy for librarians and members to use.
- **Scalability:** The database should support an increasing number of books, members, and transactions without affecting performance.

5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

Attribute	Data Type	Description
member_id	INT (Primary Key)	Unique identifier for each member
full_name	VARCHAR(100)	Member's full name
email	VARCHAR(100)	Unique email used for login and communication
phone_number	VARCHAR(15)	Member contact number
address	VARCHAR(255)	Residential address of the member
membership_date	DATE	Date the member joined the library

6. Assumptions and Constraints

- The system will be developed using a relational database (e.g., MySQL, Oracle, or PostgreSQL).
- Only registered members can borrow books.
- Each issued book must be linked to exactly one registered member at a time.

- Members must return books before the due date; otherwise, a fine will be applied.
- This document covers Week 1 requirements only. Additional modules such as Book Management, Borrowing Transactions, Fine Management, and Report Generation will be covered in the following weeks..

7. Conclusion

This document captures the initial business and functional requirements for the Library Management Database System, with specific focus on the Member Database Module. It serves as the foundation for designing the ER diagram, database schema, and subsequent implementation in the upcoming weeks.